

Matrix Theory

Being able to effectively communicate your thought process is an invaluable skill in almost any field. This course will give you plenty of opportunity to practice your mathematical writing in particular, but many of the ideas presented below apply more broadly beyond this class! Here are some tips and guidelines to keep in mind as you write up solutions and proofs for group activities, assignments, and exams in this course:

- **Think of mathematical writing in the same way you would composing an essay.**

When writing a proof or solving a mathematical problem, you likely will not come out with a clean, detailed write up right off the bat. This is completely normal, though depending on your background it may be a new sort of problem solving experience for you. It may be helpful to think of proof writing as similar to essay writing - as a process consisting of multiple parts (e.g., brainstorming, researching, outlining, drafting, and polishing).

When approaching a mathematical proof, you often have your thesis statement (the statement you are trying to prove). Figuring out the right way to approach the proof may take multiple attempts, so make sure to have scratch paper available! Your scratchwork can be as messy as it needs to be, but your final product should look polished. Sketch the problem out first, then once you have completely solved it write it up nicely on a separate sheet of paper or type up your solution on the computer.

Writing up your solution after you already know your entire approach makes for a more clear experience for the reader. For example, if you are asked to prove or disprove something, you can start your solution with a clear statement of your intent (e.g., "We will prove that the statement is true." or "We will show that the statement is not always true by giving a counterexample."). You can also make your solution more concise and to the point by identifying excess information that you did not end up using in your solution (e.g., facts you considered while solving the problem that are not used in your final argument, variables that you introduced but never used, etc.) and leave them out of your final write-up.

Make sure your conclusion is clear by finishing with a sentence tying your explanation back to the problem statement (e.g., "Thus, we have proven that if A , then B as desired." or "Thus, we have found that x must be y "). This also gives you a chance to double check that you did indeed address the prompt!

- **Keep your audience in mind**

What counts as good mathematical writing will vary depending on the context. An overview written for a layperson should look very different from an article intended for subject matter experts!



For assignments in this class, imagine you are writing for a peer who is familiar with the topics (i.e., they have watched the lecture videos and done the reading for the current module) but who has not seen the problem before. Such a reader should be able to follow your work without needing to pull out scratch paper themselves to try to figure out how you got from one step to the next. This should help you judge how much detail to include.

For this course, the other reader to be aware of is the grader. The assignments are intended as a way for you to convey your understanding of the course material. This means you need to include enough detail for the grader to be able to follow your thought process and evaluate your knowledge. If you omit steps of your process or parts of your explanation, we cannot tell if you understood them or not!

As a final note, future you is also a reader worth considering. If your solutions are clear and detailed, they will make better reference material in the future, whether you are reviewing them in this course or beyond!

When in doubt, it is better to err on the side of saying too much to make sure your solution is as clear as possible.

- **Use appropriate language and notation.**

Mathematics is in many ways its own language with its own vocabulary and notation. Your goal should be to become fluent in matrix theory over the course of this semester!

Part of this is ensuring that you use relevant terminology and notation correctly and consistently based on context. You may be familiar with alternate definitions or notations from prior linear algebra exposure, but for the sake of clarity and to avoid circular logic, you are required to use notation and definitions as presented in lecture and the textbook for this course. There is a Notation Reference Sheet provided for your convenience on the Reference Materials page on the course Canvas site.

If a problem statement introduces particular variables, you should use those variables. If you need to introduce new variables, clearly state what they represent before using them. Be careful not to use the same variable to mean multiple different things in the same solution.

- **Properly cite your sources.**

Giving credit is always important, and while you are expected to compose your own solutions in this course, your work will rely on results presented in lectures and the textbook.

For this course, you may use results/formulas/algorithms that appear in lecture videos, previous assignment problems, or the textbook without proof as long as you state that you are applying that result, note where it came from, and explain how you are using it in context. You may not use results that do not appear in either the textbook or lectures.



For theorems and definitions that appear numbered in the textbook, cite them by number (e.g., “We know from Theorem 1.2 that if x , then y .”). For results that are not numbered, make a note of where they came from and what they say (e.g., “Since x , we know from the result on Slide 2 of Lecture 1A that y ” or “We see on page 34 of Nair & Singh that...”).

You are welcome to collaborate with your peers on assignments including discussing how to approach problems, but your write up needs to be your own work that represents your individual understanding of the problem. If you work with someone, you must credit them appropriately with a note on your assignment (e.g., “Collaborated with Jane Doe”). Likewise, if you consulted any other resource (e.g., other textbooks), you should cite them as well. Note that there are restrictions on what other resources you are permitted to use in this course; please see the syllabus for more detail.

For examples of good mathematical writing in the context of our course, you can review the solutions posted for each assignment and the proofs presented in the lectures for the first couple modules.

Another nice reference on general mathematical writing is [Guidelines for Good Mathematical Writing](#) by Francis Edward Su.

If you have any questions, please let me know!

